

MoveInSync Hackathon Problem Statement

1. Problem Statement Title

Agentic Intelligence & Reporting Layer for Enterprise Mobility

2. Theme

Agentic AI (domain: Enterprise Mobility / Operations Intelligence)

3. Background / Business Context

Large enterprises move a few hundred to a few thousand employees daily through a mix of home and nodal pick-and-drop cabs and fixed-route shuttles. Transport managers are accountable for cost, safety, experience, and sustainability — but most of their time goes into assembling data, not acting on it. The signal is rich. The insight is missing. The actions are manual.

Three personas feel this differently:

- **Transport manager** (operational) — owns day-to-day ops: vendor coordination, escalations, shift planning, delay management. Needs fast, actionable signals, not reports.
- **Transport & facilities head** (strategic) — owns budget, SLA accountability, vendor strategy, leadership reporting. Needs a coherent cost/safety/experience story without assembling it manually.
- **Team / line manager** (shift-based ops) — needs shift-level visibility into who made it, who was late, and how delays ripple into floor/ops readiness.

4. Problem Definition

Mobility operations generate rich structured data continuously (trips, vendors, drivers, employees — ops scale, timeliness/delays, safety/compliance, cost, sustainability, employee experience, vendor performance). Today this largely sits in static weekly/monthly reports.

Pain point: a metric without context is just a number. E.g. "OTA is 78%" matters far less than "it was 85% last month, SLA is 90%, and two vendors are responsible for the gap." Benchmarking against historical trend / SLA / industry norm / peer is currently absent.

Desired improvement: an agentic layer that senses what's happening, reasons about what it means, and acts — with minimal human prompting — surfacing what matters, benchmarking it, and communicating outcomes to the right person at the right time.

5. Tech Stack

Open / participant's choice — *preferably Java, Angular, AWS resources, but not restrictive.*

6. Resources Provided

Sample dataset only: anonymised trip logs across cab, nodal, and shuttle modes — including vendor performance, GPS traces, delay records, cost data, and employee feedback.

7. Expected Output

Solutions may take any of the following forms, or combine several:

- Conversational agent (NL Q&A on mobility data)
- Proactive alerting & triggers
- Automated reporting & narratives
- Insight & anomaly detection
- Decision-support dashboard
- Automated communications

8. Requirements

Mandatory

- Working, demo-able prototype running on the provided dataset
- Agentic behaviour — senses, reasons, and acts; not a passive dashboard or query-only tool
- Serves at least one of the three named personas
- Contextualises metrics against at least one reference point (historical trend, SLA/goal, industry benchmark, or peer comparison)

Good-to-have

- Combines two or more of the solution forms listed in Section 7
- Handles messy or missing data gracefully (GPS gaps, unmatched records, incomplete rosters)
- Proactive triggers rather than purely on-demand responses

Bonus

- Credible deployability story into an existing enterprise mobility platform — multi-tenancy, latency, cost
- Output a transport & facilities head could forward to leadership without rework

Not expected

- Production-grade authentication or security
- A full historical data pipeline
- Integration with real vendor systems

Constraints

- Anonymised sample trip-log dataset only — no live system access

9. Evaluation Criteria

#	Criteria	What we are looking for	Weight
1	Business impact & experience	Does it meaningfully reduce manager effort or surface decisions that would otherwise be missed — and does it land? Clarity for the intended persona, leadership-ready output, shareable without rework.	35
2	Agentic design & cost at scale	Is AI solving a genuine problem rather than decorating the solution — and can it actually be run? Inference cost per interaction, latency, efficiency at enterprise volumes.	20

#	Criteria	What we are looking for	Weight
3	Architecture & code quality	Sound structure and engineering judgement; deployable into an existing platform; choices a team could build on.	20
4	Functionality	It runs — a working, demo-able prototype, end to end, on the provided dataset.	25
	Total		100

10. Deliverables

- Source code repository (GitHub/GitLab)
- Architecture diagram
- README + setup instructions
- Sample inputs/outputs
- Demo video (if requested)
- Presentation deck
- Live demo